

# Mazen Mohamed Abd El Moneim Gadallah

AI & Machine Learning Enthusiast | Telecommunications Engineering Student

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## PROFILE

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AI and machine learning enthusiast studying computer engineering with a strong interest in telecommunications. Builds practical projects across machine learning, deep learning, mobile development, software systems, and embedded platforms, with a focus on turning technical ideas into useful solutions.

## EXPERIENCE & SELECTED PROJECTS

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### Mobile Security Testing Platform

AI Orchestration Layer Contributor | Present | Project Experience

- Building the AI orchestration and decision-making layer on top of an automated mobile penetration testing tool as part of a cross-functional AI team.
- Collaborating with cloud and security engineering teams to integrate AI-driven decision logic into the existing testing pipeline.

**Technologies:** Python, Artificial Intelligence, Decision Logic, Mobile Security Testing

### Digital Egypt Pioneers (DEPI) Program

Program Participant | Present | Training Program

- Produced professional project kick-off and mid-project documentation for multiple initiatives, including a chat application and SecureBot, a security-focused chatbot.
- Practiced client-facing communication through freelance proposal writing using the HURLEC framework, project documentation, and professional email correspondence.

**Technologies:** Technical Documentation, Client Communication, Chat Applications, Security-Focused AI

### AI Student Grade Prediction Model

Machine Learning Project | Personal Project

- Problem: estimate a student's expected grade from input features.
- Solution: built a prediction workflow with preprocessing, feature engineering, model training, and performance evaluation using machine learning and deep learning techniques.

**Technologies:** Python, Machine Learning, Deep Learning, Data Preprocessing, Feature Engineering

### Vesita Mobile Application

Mobile Application Project | Personal Project

- Problem: provide a simple mobile experience around the application's core features.
- Solution: built a Flutter application with a user-friendly interface and implemented the main platform features.

**Technologies:** Flutter, Dart

### To-Do List Mobile Application

Mobile Application Project | Personal Project

- Problem: organize personal tasks in one place.
- Solution: developed a task management app for creating, organizing, and managing tasks, supported by technical documentation.

**Technologies:** Flutter, Dart

### E-Commerce Systems & Backend

Backend Development Project | Personal Project

- Problem: support a structured e-commerce platform with reliable data persistence.
- Solution: built an e-commerce backend with Spring Boot, PostgreSQL, and JPA/Hibernate for maintainable system structure and persistence.

**Technologies:** Java, Spring Boot, PostgreSQL, JPA/Hibernate, Maven

### LLM & Deep Learning Research Projects

AI Research Project | Research Project

- Problem: compare language-model autocomplete quality under latency constraints.
- Solution: benchmarked DistilGPT-2, GPT-2, and GPT-2 Medium, explored temperature, top-k, top-p, and penalty decoding, and implemented retrieval-augmented generation pipelines.

**Technologies:** Python, DistilGPT-2, GPT-2, RAG, Hugging Face

Automated Document Generation Pipeline

Automation Project | Personal Project

- Problem: generate consistent Word and PDF documents without repetitive manual formatting.
- Solution: built an automated document generation and conversion pipeline with Node.js, docx, and LibreOffice.

Technologies: Node.js, docx, LibreOffice, PDF Conversion

EDUCATION

B.Sc. in Computer Engineering - AI & Software Development

El Shorouk Academy (Modern Academy) | Present

Cairo, Egypt. Degree in progress.

CERTIFICATIONS & COURSES

|                                                               |                                                                      |
|---------------------------------------------------------------|----------------------------------------------------------------------|
| CCNA: Introduction to Networks                                | Cisco Networking Academy through Modern Academy   2025/07            |
| Python Programming Basics                                     | ITI Platform Mahara-Tech - Artificial Intelligence Academy   2026/01 |
| Python Programming Certificate                                | Ministry of Communications and Information Technology (MCIT)         |
| Huawei HCIA-Cloud Computing V5.5 Training                     | Huawei ICT Academy                                                   |
| Oracle Cloud Infrastructure (OCI) AI Certification Coursework | Oracle                                                               |
| Machine Learning & Deep Learning Certificate                  |                                                                      |
| Flutter Development Certificate                               |                                                                      |
| Software Systems Courses                                      |                                                                      |

SKILLS

|                               |                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages         | Python   C++   C   Java   Dart   JavaScript / Node.js   SQL   MATLAB                                                                |
| AI & Machine Learning         | Machine Learning   Deep Learning   AI Model Development   LLM Architecture & Decoding   Hugging Face   Scikit-Learn   RAG Pipelines |
| Embedded & Telecommunications | Arduino / ESP32   Telecommunications   CCNA   Computer Networking   Subnetting and VLSM   TCP / UDP                                 |
| Mobile & Software Systems     | Flutter   Spring Boot   PostgreSQL   JPA / Hibernate   Maven   System Design   Software Architecture                                |
| Tools                         | Git   GitHub   Docker   Linux   LibreOffice                                                                                         |
| Professional Skills           | Problem Solving   Object-Oriented Programming   Analytical Thinking   Teamwork   Technical Documentation   Fast Learning            |

LANGUAGES

Arabic: Native | English: Professional Working Proficiency